

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of B.Sc. Biology Examination March-April 2018

BS303 Concepts of Physics-II

Date: 02-04-2018, Monday

Time: 10.00 AM To 10.20 AM

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-1 Choose the correct answer for the following questions [10]

1. CGS unit for the magnetic field is _____.
a) Tesla
b) Gauss
c) Joule / C
d) N / C
1
2. _____ is measure of the bending of a ray of light when passing from one medium into another.
a) the index of refraction
b) the diffraction
c) the angle of reflection
d) refraction
1
3. The _____ relates the force between two single charges separated by a square of distance.
a) Ohm's law
b) Faraday's law
c) Coulomb's law
d) Ampere's law
1
4. A _____ has an enough binding energy to hold the nucleus together permanently.
a) Stable Atom
b) Unstable Atom
c) Excited Atom
d) None
1
5. The maximum ($\theta = \pi/2$) magnetic torque can be written as _____.
a) BA
b) BI
c) BN
d) $B\mu$
1
6. In _____, the focal length (f) is negative.
a) concave lens and concave mirror
b) convex lens and convex mirror
c) convex lens and concave mirror
d) concave lens and convex mirror
1

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III Semester of B.Sc. Biology Examination March-April 2018

BS303 Concepts of Physics - II

Date: 02-04-2018, Monday

Time: 10.20 AM To 12.00 PM

Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed (Any Five) 10

1. Discuss Faraday's law of induction. 2
2. What is the difference between Specular and Diffuse reflection? 2
3. How to distinguish a stable atom and an unstable atom? 2
4. Explain about the membrane work as capacitors. 2
5. Define dispersion and diffraction. 2
6. Write the relation among the voltage, current and resistance and its physical significance. 2
7. If charged parallel plates are held at a potential difference of 200 Volts, how much work does it take to move one electron from the positive plate to the negative plate? (charge of the electron = $1.6 \times 10^{-19} C$) 2

SECTION - II

Q-III Answer the following questions as directed 15

1. What is the principle of conservation of charge? Draw the field lines for individual-positive and negative charges, two like and two unlike charges. 3
OR
Write a note on Michelson interferometer. 3
2. Discuss on self-inductance. 3
3. Explain the radioactivity. 3
OR
(a) What is magnification? (b) Write down Mirror equation. (c) Give any two applications of Fibre Optics. 3
4. Write law of refraction and reflection with proper diagram. Discuss on refractive index. 3
5. Give the representation of polarized light and unpolarized light. Discuss on the Polarization. 3
OR
Discuss the optical activity with proper diagram. 3